

$$y_t = (\phi_1 y_{t-1} + \phi_2 y_{t-2} + \phi_3 y_{t-3} + e_t - \theta_1 e_{t-1} - \theta_2 e_{t-2} - \theta_3 e_{t-3})$$

$e_t \sim \text{N}(0, \sigma_e^2)$

$$E(y_t) = 0 //$$

$$\gamma_k = \text{cov}(y_t, y_{t-k})$$

$$= \text{cov}(\phi_1 y_{t-1} + \phi_2 y_{t-2} + \phi_3 y_{t-3} + e_t - \theta_1 e_{t-1} - \theta_2 e_{t-2} - \theta_3 e_{t-3}, y_{t-k})$$

$$= \phi_1 \gamma_{k-1} + \phi_2 \gamma_{k-2} + \phi_3 \gamma_{k-3} + \text{cov}(e_t, y_{t-k}) - \theta_1 \text{cov}(e_{t-1}, y_{t-k}) - \theta_2 \text{cov}(e_{t-2}, y_{t-k}) - \theta_3 \text{cov}(e_{t-3}, y_{t-k})$$

$$\begin{aligned} \gamma_0 &= \phi_1 \gamma_1 + \phi_2 \gamma_2 + \phi_3 \gamma_3 + \sigma_e^2 - \theta_1 (\phi_1 \sigma_e^2 - \theta_1 \sigma_e^2) - \theta_2 (\phi_2 \sigma_e^2 - \theta_2 \sigma_e^2) - \theta_3 (\phi_3 \sigma_e^2 - \theta_3 \sigma_e^2) \\ &= \phi_1 \gamma_1 + \phi_2 \gamma_2 + \phi_3 \gamma_3 + \sigma_e^2 (1 - \theta_1 (\phi_1 - \theta_1) - \theta_2 (\phi_2 - \theta_2) - \theta_3 (\phi_3 - \theta_3)) \end{aligned}$$

$$\begin{aligned} \gamma_1 &= \phi_1 \gamma_0 + \phi_2 \gamma_1 + \phi_3 \gamma_2 + 0 - \theta_1 \sigma_e^2 - \theta_2 (\phi_1 \sigma_e^2 - \theta_1 \sigma_e^2) - \theta_3 (\phi_2 \sigma_e^2 - \theta_2 \sigma_e^2) \\ &= \phi_1 \gamma_0 + \phi_2 \gamma_1 + \phi_3 \gamma_2 - \sigma_e^2 (-\theta_1 - \theta_2 (\phi_1 - \theta_1) - \theta_3 (\phi_2 - \theta_2)) \end{aligned}$$

$$\gamma_1 - \phi_2 \gamma_1 = \quad - \parallel \quad \underline{\hspace{1cm}}$$

$$\rho_k = \frac{\gamma_k}{\gamma_0} \quad \rho_1 = \frac{\gamma_1}{\gamma_0}$$

$$(1 - \phi_2) \gamma_1 = \quad - \parallel \quad \underline{\hspace{1cm}}$$

$$\gamma_1 =$$